

Solutions

Chapter 1 · All questions with full solutions



Part 1 · Examples

Q

1.1

Mole fraction
from mass
percent



Calculate the mole fraction of ethylene glycol ($\text{C}_2 \text{H}_6 \text{O}_2$) in a solution containing 20 % of $\text{C}_2 \text{H}_6 \text{O}_2$ by mass.

Given

Assume we have 100 g of solution — we can start with any amount, since the result will be the same.

This solution contains 20 g of ethylene glycol and 80 g of water.

Substitute

Molar mass of $\text{C}_2 \text{H}_6 \text{O}_2 = 12 \times 2 + 1 \times 6 + 16 \times 2 = 62 \text{ g mol}^{-1}$.

$$\text{Moles of } \text{C}_2 \text{H}_6 \text{O}_2 = \frac{20 \text{ g}}{62 \text{ g mol}^{-1}} = 0.322 \text{ mol}$$

$$\text{Moles of water} = \frac{80 \text{ g}}{18 \text{ g mol}^{-1}} = 4.444 \text{ mol}$$

$$\begin{aligned} x_{\text{glycol}} &= \frac{\text{moles of } \text{C}_2 \text{H}_6 \text{O}_2}{\text{moles of } \text{C}_2 \text{H}_6 \text{O}_2 + \text{moles of } \text{H}_2 \text{O}} \\ &= \frac{0.322 \text{ mol}}{0.322 \text{ mol} + 4.444 \text{ mol}} = 0.068 \end{aligned}$$

Conclusion

$$\text{Similarly, } x_{\text{water}} = \frac{4.444 \text{ mol}}{0.322 \text{ mol} + 4.444 \text{ mol}} = 0.932$$

Mole fraction of water can also be calculated as: $1 - 0.068 = 0.932$

$$x_{\text{glycol}} = 0.068 \text{ (and } x_{\text{water}} = 0.932) //$$

Q

1.2

Molarity
from mass
and volume



Calculate the molarity of a solution containing 5 g of NaOH in 450 mL solution.

Conclusion

$$\text{Moles of NaOH} = \frac{5 \text{ g}}{40 \text{ g mol}^{-1}} = 0.125 \text{ mol}$$

$$\text{Volume of the solution in litres} = \frac{450 \text{ mL}}{1000 \text{ mL L}^{-1}}$$

Using equation (2.8),

$$\text{Molarity} = \frac{0.125 \text{ mol} \times 1000 \text{ mL L}^{-1}}{450 \text{ mL}} = 0.278 \text{ M}$$

$$= 0.278 \text{ mol L}^{-1}$$

$$= 0.278 \text{ mol dm}^{-3}$$

$$0.278 \text{ M (} 0.278 \text{ mol L}^{-1} \text{)} //$$

Q

1.3

Molality of ethanoic acid in benzene

()

Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75 g of benzene.

Conclusion

Molar mass of $\text{C}_2\text{H}_4\text{O}_2$: $12 \times 2 + 1 \times 4 + 16 \times 2 = 60 \text{ g mol}^{-1}$

Solution

$$\text{Moles of } \text{C}_2\text{H}_4\text{O}_2 = \frac{2.5 \text{ g}}{60 \text{ g mol}^{-1}} = 0.0417 \text{ mol}$$

$$\text{Mass of benzene in kg} = \frac{75 \text{ g}}{1000 \text{ g kg}^{-1}} = 75 \times 10^{-3} \text{ kg}$$

$$\begin{aligned} \text{Molality of } \text{C}_2\text{H}_4\text{O}_2 &= \frac{\text{Moles of } \text{C}_2\text{H}_4\text{O}_2}{\text{kg of benzene}} = \frac{0.0417 \text{ mol} \times 1000 \text{ g kg}^{-1}}{75 \text{ g}} \\ &= 0.556 \text{ mol kg}^{-1} \end{aligned}$$

$$0.556 \text{ mol kg}^{-1} //$$

Q

1.4

Henry's law solubility of nitrogen in water



If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N_2 at 293 K is 76.48 kbar.

Given

The solubility of a gas is related to its mole fraction in the aqueous solution, which we find using Henry's law.

Thus:

$$x(\text{Nitrogen}) = \frac{p(\text{nitrogen})}{K_H} = \frac{0.987 \text{ bar}}{76,480 \text{ bar}} = 1.29 \times 10^{-5}$$

1 litre of water contains 55.5 mol of water. If n represents number of moles of N_2 in solution,

$$x(\text{Nitrogen}) = \frac{n \text{ mol}}{n \text{ mol} + 55.5 \text{ mol}} = \frac{n}{55.5} = 1.29 \times 10^{-5}$$

Tip

(n in denominator is neglected as it is $\ll 55.5$)

Conclusion

$$\begin{aligned} \text{Thus } n &= 1.29 \times 10^{-5} \times 55.5 \text{ mol} = 7.16 \times 10^{-4} \text{ mol} \\ &= \frac{7.16 \times 10^{-4} \text{ mol} \times 1000 \text{ mmol}}{1 \text{ mol}} = 0.716 \text{ mmol} \end{aligned} //$$

Q

1.5

Vapour pressure of an ideal binary solution



Vapour pressure of chloroform (CHCl_3) and dichloromethane (CH_2Cl_2) at 298 K are 200 mm Hg and 415 mm Hg respectively. (i) Calculate the vapour pressure of the solution prepared by mixing 25.5 g of CHCl_3 and 40 g of CH_2Cl_2 at 298 K and, (ii) mole fractions of each component in vapour phase.

Given

(i) Molar mass of CH_2Cl_2 = $12 \times 1 + 1 \times 2 + 35.5 \times 2 = 85 \text{ g mol}^{-1}$

Molar mass of CHCl_3 = $12 \times 1 + 1 \times 1 + 35.5 \times 3 = 119.5 \text{ g mol}^{-1}$

$$\text{Moles of } \text{CH}_2\text{Cl}_2 = \frac{40 \text{ g}}{85 \text{ g mol}^{-1}} = 0.47 \text{ mol}$$

$$\text{Moles of CHCl}_3 = \frac{25.5 \text{ g}}{119.5 \text{ g mol}^{-1}} = 0.213 \text{ mol}$$

$$\text{Total number of moles} = 0.47 \text{ mol} + 0.213 \text{ mol} = 0.683 \text{ mol}$$

$$x_{\text{CH}_2\text{Cl}_2} = \frac{0.47 \text{ mol}}{0.683 \text{ mol}} = 0.688$$

$$x_{\text{CHCl}_3} = 1.00 - 0.688 = 0.312$$

Using equation (2.16),

$$\begin{aligned} p_{\text{total}} &= p_1^0 + (p_2^0 - p_1^0) x_2 = 200 + (415 - 200) \times 0.688 \\ &= 200 + 147.9 = 347.9 \text{ mm Hg} \end{aligned}$$

Conclusion

(ii) Using relation (2.19), $y_i = p_i / p_{\text{total}}$, we get the mole fraction of each component in the gas phase (y_i).

$$p_{\text{CH}_2\text{Cl}_2} = 0.688 \times 415 \text{ mm Hg} = 285.5 \text{ mm Hg}$$

$$p_{\text{CHCl}_3} = 0.312 \times 200 \text{ mm Hg} = 62.4 \text{ mm Hg}$$

$$y_{\text{CH}_2\text{Cl}_2} = \frac{285.5 \text{ mm Hg}}{347.9 \text{ mm Hg}} = 0.82$$

$$y_{\text{CHCl}_3} = \frac{62.4 \text{ mm Hg}}{347.9 \text{ mm Hg}} = 0.18$$

$$p_{\text{total}} = 347.9 \text{ mm Hg}; y_{\text{CH}_2\text{Cl}_2} = 0.82, y_{\text{CHCl}_3} = 0.18$$

Recall

Note: Since, CH_2Cl_2 is more volatile than CHCl_3 , [$p_{\text{CH}_2\text{Cl}_2}^0 = 415 \text{ mm Hg}$ and $p_{\text{CHCl}_3}^0 = 200 \text{ mm Hg}$] and the vapour phase is also richer in CH_2Cl_2 [$y_{\text{CH}_2\text{Cl}_2} = 0.82$ and $y_{\text{CHCl}_3} = 0.18$]. So at equilibrium, the vapour phase is always richer in the more volatile component.

Q

1.6

Molar mass from vapour pressure lowering



The vapour pressure of pure benzene at a certain temperature is 0.850 bar. A non-volatile, non-electrolyte solid weighing 0.5 g when added to 39.0 g of benzene (molar mass 78 g mol^{-1}). Vapour pressure of the solution, then, is 0.845 bar. What is the molar mass of the solid substance?

Given

The known quantities are:

$$p_1^0 = 0.850 \text{ bar}; p = 0.845 \text{ bar}; M_1 = 78 \text{ g mol}^{-1}; w_2 = 0.5 \text{ g}; w_1 = 39 \text{ g}$$

Substitute

Substituting these values into equation (2.28), we get:

$$\frac{0.850 \text{ bar} - 0.845 \text{ bar}}{0.850 \text{ bar}} = \frac{0.5 \text{ g} \times 78 \text{ g mol}^{-1}}{M_2 \times 39 \text{ g}}$$

Conclusion

$$\text{So } M_2 = 170 \text{ g mol}^{-1} \quad \boxed{170 \text{ g mol}^{-1}}$$

Q

1.7

Boiling point
elevation of a
glucose
solution

()

18 g of glucose, $C_6H_{12}O_6$, is dissolved in 1 kg of water in a saucepan. At what temperature will water boil at 1.013 bar? K_b for water is $0.52 \text{ K kg mol}^{-1}$.

Substitute

$$\text{Moles of glucose} = \frac{18 \text{ g}}{180 \text{ g mol}^{-1}} = 0.1 \text{ mol}$$

$$\text{Number of kilograms of solvent} = 1 \text{ kg}$$

$$\text{Thus molality of glucose solution} = 0.1 \text{ mol kg}^{-1}$$

$$\text{For water, change in boiling point } \Delta T_b = K_b \times m = 0.52 \text{ K kg mol}^{-1} \times 0.1 \text{ mol kg}^{-1} = 0.052 \text{ K}$$

Conclusion

Water boils at 373.15 K at 1.013 bar pressure. So the boiling point of the solution is $373.15 + 0.052 = 373.202 \text{ K}$.

$$373.202 \text{ K} //$$

Q

1.8

Molar mass
from boiling
point
elevation



The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile solute was dissolved in 90 g of benzene, the boiling point is raised to 354.11 K. Calculate the molar mass of the solute. K_b for benzene is $2.53 \text{ K kg mol}^{-1}$

Substitute

The elevation (ΔT_b) in the boiling point = $354.11 \text{ K} - 353.23 \text{ K} = 0.88 \text{ K}$. Substituting these values into expression (2.33), we get:

$$M_2 = \frac{2.53 \text{ K kg mol}^{-1} \times 1.8 \text{ g} \times 1000 \text{ g kg}^{-1}}{0.88 \text{ K} \times 90 \text{ g}} = 58 \text{ g mol}^{-1}$$

Conclusion

So the molar mass of the solute is $M_2 = 58 \text{ g mol}^{-1}$. $58 \text{ g mol}^{-1} //$

Q

1.9

Freezing
point
depression of
ethylene
glycol
solution



45 g of ethylene glycol ($C_2H_6O_2$) is mixed with 600 g of water. Calculate (a) the freezing point depression and (b) the freezing point of the solution.

Substitute

Depression in freezing point depends on molality. So the molality of the solution with respect to ethylene glycol = $\frac{\text{moles of ethylene glycol}}{\text{mass of water in kilogram}}$

$$\text{Moles of ethylene glycol} = \frac{45 \text{ g}}{62 \text{ g mol}^{-1}} = 0.73 \text{ mol}$$

$$\text{Mass of water in kg} = \frac{600 \text{ g}}{1000 \text{ g kg}^{-1}} = 0.6 \text{ kg}$$

$$\text{Hence molality of ethylene glycol} = \frac{0.73 \text{ mol}}{0.60 \text{ kg}} = 1.2 \text{ mol kg}^{-1}$$

So the freezing point depression is:

$$\Delta T_f = 1.86 \text{ K kg mol}^{-1} \times 1.2 \text{ mol kg}^{-1} = 2.2 \text{ K}$$

Conclusion

Freezing point of the aqueous solution = $273.15 \text{ K} - 2.2 \text{ K} = 270.95 \text{ K}$

$$\Delta T_f = 2.2 \text{ K; freezing point of solution} = 270.95 \text{ K} //$$

Q

1.10

Molar mass
from
freezing
point
depression



1.00 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is 5.12 K kg mol⁻¹. Find the molar mass of the solute.

Substitute

Substituting the values of various terms involved in equation (1.36) we get,

$$M_2 = \frac{5.12 \text{ K kg mol}^{-1} \times 1.00 \text{ g} \times 1000 \text{ g kg}^{-1}}{0.40 \times 50 \text{ g}} = 256 \text{ g mol}^{-1}$$

Conclusion

Thus, molar mass of the solute = 256 g mol⁻¹ **256 g mol⁻¹ //**

Q

1.11

Molar mass
of a protein
from osmotic
pressure

()

200 cm³ of an aqueous solution of a protein contains 1.26 g of the protein. The osmotic pressure of such a solution at 300 K is found to be 2.57 × 10⁻³ bar. Calculate the molar mass of the protein.

Given

The known quantities are: $\Pi = 2.57 \times 10^{-3}$ bar,

$V = 200 \text{ cm}^3 = 0.200$ litre

$T = 300$ K

$R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$

Substitute

Substituting these values into equation (2.42), we get:

$$M_2 = \frac{1.26 \text{ g} \times 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}}{2.57 \times 10^{-3} \text{ bar} \times 0.200 \text{ L}} = 61,022 \text{ g mol}^{-1}$$

61,022 g mol⁻¹ //

Q

1.12

Association
of benzoic
acid in
benzene



2 g of benzoic acid (C₆ H₅ COOH) dissolved in 25 g of benzene shows a depression in freezing point equal to 1.62 K. Molal depression constant for benzene is 4.9 K kg mol⁻¹. What is the percentage association of acid if it forms dimer in solution?

Given

The known quantities are: $w_2 = 2 \text{ g}$; $K_f = 4.9 \text{ K kg mol}^{-1}$; $w_1 = 25 \text{ g}$,

$\Delta T_f = 1.62 \text{ K}$

Substitute

Substituting these values into equation (1.36), we get:

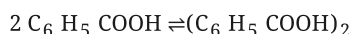
$$M_2 = \frac{4.9 \text{ K kg mol}^{-1} \times 2 \text{ g} \times 1000 \text{ g kg}^{-1}}{25 \text{ g} \times 1.62 \text{ K}} = 241.98 \text{ g mol}^{-1}$$

So the experimental molar mass of benzoic acid in benzene is

= 241.98 g mol⁻¹

Given

Now consider this equilibrium for the acid:



Let x be the degree of association of the solute. Then $(1-x)$ mol of benzoic acid remains unassociated, and $\frac{x}{2}$ mol is present as associated benzoic acid at

equilibrium. So the total number of moles of particles at equilibrium is:

$$1-x + \frac{x}{2} = 1 - \frac{x}{2}$$

So the total number of moles of particles at equilibrium equals the van't Hoff factor i .

Key formula

$$\text{But } i = \frac{\text{Normal molar mass}}{\text{Abnormal molar mass}}$$



Substitute

$$= \frac{122 \text{ g mol}^{-1}}{241.98 \text{ g mol}^{-1}}$$

or

$$\frac{x}{2} = 1 - \frac{122}{241.98} = 1 - 0.504 = 0.496$$

or

$$x = 2 \times 0.496 = 0.992$$

Conclusion

So the degree of association of benzoic acid in benzene is 99.2 %.

$$\text{Degree of association} = 99.2\%$$

Q

1.13

Van't Hoff factor and dissociation constant of acetic acid



0.6 mL of acetic acid (CH_3COOH), having density 1.06 g mL^{-1} , is dissolved in 1 litre of water. The depression in freezing point observed for this strength of acid was 0.0205°C . Calculate the van't Hoff factor and the dissociation constant of acid.

Substitute

$$\text{Number of moles of acetic acid} = \frac{0.6 \text{ mL} \times 1.06 \text{ g mL}^{-1}}{60 \text{ g mol}^{-1}}$$

$$= 0.0106 \text{ mol} = n$$

$$\text{Molality} = \frac{0.0106 \text{ mol}}{1000 \text{ mL} \times 1 \text{ g mL}^{-1}} = 0.0106 \text{ mol kg}^{-1}$$

Using equation (1.35)

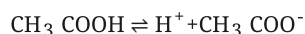
$$\Delta T_f = 1.86 \text{ K kg mol}^{-1} \times 0.0106 \text{ mol kg}^{-1} = 0.0197 \text{ K}$$

$$\text{van't Hoff Factor (i)} = \frac{\text{Observed freezing point}}{\text{Calculated freezing point}} = \frac{0.0205 \text{ K}}{0.0197 \text{ K}} = 1.041$$

Given



Acetic acid is a weak electrolyte. It dissociates into two ions per molecule: acetate and hydrogen ions. Let x be the degree of dissociation of acetic acid. Then we have $n(1-x)$ moles of undissociated acetic acid, $n x$ moles of CH_3COO^- and $n x$ moles of H^+ ions,



$$n \text{ mol } 0 \quad 0$$

$$n(1-x) \quad n x \text{ mol } n x \text{ mol}$$

Thus total moles of particles are: $n(1-x+x+x) = n(1+x)$

Substitute

$$i = \frac{n(1+x)}{n} = 1+x = 1.041$$

$$\text{Thus degree of dissociation of acetic acid} = x = 1.041 - 1.000 = 0.041$$

Conclusion

Then $[\text{CH}_3 \text{ COOH}] = n(1-x) = 0.0106(1-0.041)$,

$[\text{CH}_3 \text{ COO}^-] = n x = 0.0106 \times 0.041$, $[\text{H}^+] = n x = 0.0106 \times 0.041$.

$$K_a = \frac{[\text{CH}_3 \text{ COO}^-][\text{H}^+]}{[\text{CH}_3 \text{ COOH}]} = \frac{0.0106 \times 0.041 \times 0.0106 \times 0.041}{0.0106(1.00-0.041)}$$

$$= 1.86 \times 10^{-5}$$

$$\mathbf{i = 1.041; K_a = 1.86 \times 10^{-5} //}$$

Part 2 · Additional Questions

Q

1

Mass
percentage
from mass of
solute / solvent



Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) if 22 g of benzene is dissolved in 122 g of carbon tetrachloride.

Key formula



$$\text{Mass percentage of } C_6H_6 = \frac{\text{Mass of } C_6H_6}{\text{Total mass of the solution}} \times 100\%$$

Substitute

$$= \frac{\text{Mass of } C_6H_6}{\text{Mass of } C_6H_6 + \text{Mass of } CCl_4} \times 100\%$$

$$= \frac{22}{22+122} \times 100\%$$

$$= 15.28\%$$

$$= \frac{\text{Mass of } CCl_4}{\text{Total mass of the solution}} \times 100\%$$

Mass percentage of CCl_4

$$= \frac{\text{Mass of } CCl_4}{\text{Mass of } C_6H_6 + \text{Mass of } CCl_4} \times 100\%$$

$$= \frac{122}{22+122} \times 100\%$$

$$= 84.72\%$$

Conclusion

Alternatively,

Mass percentage of $CCl_4 = (100-15.28)\%$

$$= 84.72\%$$

Mass percentage of $C_6H_6 = 15.28\%$; Mass percentage of $CCl_4 = 84.72\%$



Q

2

Mole fraction
of benzene in
 CCl_4

(())

Calculate the mole fraction of benzene in solution containing 30% by mass in carbon tetrachloride.

Given



Let the total mass of the solution be 100 g and the mass of benzene be 30 g.

$\therefore$ Mass of carbon tetrachloride = (100-30) g

$$= 70 \text{ g}$$

Substitute

Molar mass of benzene (C_6H_6) = $(6 \times 12 + 6 \times 1) \text{ g mol}^{-1}$

$$= 78 \text{ g mol}^{-1}$$

$$\therefore \text{Number of moles of } C_6H_6 = \frac{30}{78} \text{ mol}$$

$$= 0.3846 \text{ mol}$$

Molar mass of carbon tetrachloride (CCl_4) = $1 \times 12 + 4 \times 35.5$

$$= 154 \text{ g mol}^{-1}$$

$$\therefore \text{Number of moles of } \text{CCl}_4 = \frac{70}{154} \text{ mol}$$

$$= 0.4545 \text{ mol}$$

Key formula

Thus, the mole fraction of C_6H_6 is given as:



$$\frac{\text{Number of moles of } \text{C}_6\text{H}_6}{\text{Number of moles of } \text{C}_6\text{H}_6 + \text{Number of moles of } \text{CCl}_4}$$

Conclusion

$$= \frac{0.3846}{0.3846 + 0.4545}$$

$$= 0.458$$

Mole fraction of C_6H_6 = 0.458 //

Q

3

Molarity of
two solutions



Calculate the molarity of each of the following solutions: (a) 30 g of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 4.3 L of solution (b) 30 mL of 0.5 M H_2SO_4 diluted to 500 mL.

Key formula

Molarity is given by:



$$\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}}$$

Substitute

(a) Molar mass of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ = $59 + 2(14 + 3 \times 16) + 6 \times 18$

$$= 291 \text{ g mol}^{-1}$$

$$\therefore \text{Moles of } \text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} = \frac{30}{291} \text{ mol}$$

$$= 0.103 \text{ mol}$$

$$\text{Therefore, molarity} = \frac{0.103 \text{ mol}}{4.3 \text{ L}}$$

$$= 0.023 \text{ M}$$

(b) Number of moles present in 1000 mL of 0.5 M H_2SO_4 = 0.5 mol

$$\therefore \text{Number of moles present in 30 mL of 0.5 M } \text{H}_2\text{SO}_4 = \frac{0.5 \times 30}{1000} \text{ mol}$$

$$= 0.015 \text{ mol}$$

$$\text{Therefore, molarity} = \frac{0.015 \text{ mol}}{0.5 \text{ L}}$$

$$= 0.03 \text{ M}$$

(a) 0.023 M; (b) 0.03 M //

Q

4

Mass of urea
for a molal
solution



Calculate the mass of urea (NH_2CONH_2) required in making 2.5 kg of 0.25 molal aqueous solution.

Given

$$\begin{aligned}\text{Molar mass of urea (NH}_2\text{CONH}_2) &= 2(1 \times 14 + 2 \times 1) + 1 \times 12 + 1 \times 16 \\ &= 60 \text{ g mol}^{-1}\end{aligned}$$

0.25 molal aqueous solution of urea means:

$$\begin{aligned}1000 \text{ g of water contains } 0.25 \text{ mol} &= (0.25 \times 60) \text{ g of urea} \\ &= 15 \text{ g of urea}\end{aligned}$$

That is,

(1000+15) g of solution contains 15 g of urea

Substitute

Therefore, 2.5 kg(2500 g) of solution contains

$$\begin{aligned}&= \frac{15 \times 2500}{1000+15} \text{ g} \\ &= 36.95 \text{ g} \\ &= 37 \text{ g of urea (approximately)}\end{aligned}$$

Conclusion

Hence, mass of urea required = 37 g

Note: There is a slight variation in this answer and the one given in the NCERT textbook.

37 g of urea //

Q

5

Molality / mo-
larity / mole
fraction of KI
solution



Calculate (a) molality (b) molarity and (c) mole fraction of KI if the density of 20% (mass / mass) aqueous KI is 1.202 g mL^{-1} .

Substitute

$$\text{(a) Molar mass of KI} = 39 + 127 = 166 \text{ g mol}^{-1}$$

20% (mass / mass) aqueous solution of KI means 20 g of KI is present in 100 g of solution.

That is,

20 g of KI is present in (100-20) g of water = 80 g of water

$$\begin{aligned}&= \frac{\text{Moles of KI}}{\text{Mass of water in kg}} \\ &= \frac{\frac{20}{166}}{0.08} \text{ m}\end{aligned}$$

$$= 1.506 \text{ m}$$

$$= 1.51 \text{ m (approximately)}$$

(b) It is given that the density of the solution = 1.202 g mL^{-1}

$$\therefore \text{Volume of 100 g solution} = \frac{\text{Mass}}{\text{Density}}$$

$$= \frac{100 \text{ g}}{1.202 \text{ g mL}^{-1}}$$

$$= 83.19 \text{ mL}$$

$$= 83.19 \times 10^{-3} \text{ L}$$

Therefore, molarity of the solution

$$= \frac{\frac{20}{166} \text{ mol}}{83.19 \times 10^{-3} \text{ L}}$$

$$= 1.45 \text{ M}$$

(c) Moles of KI

Moles of water

$$\text{Therefore, } = \frac{20}{166} = 0.12 \text{ mol}$$

$$= 0.0263 = \frac{80}{18} = 4.44 \text{ mol}$$

mole fraction of KI

$$= \frac{0.12}{0.12+4.44}$$

$$= \frac{\text{Moles of KI}}{\text{Moles of KI} + \text{Moles of water}}$$

(a) 1.51 m; (b) 1.45 M; (c) mole fraction of KI = 0.0263 //

Q

6

Henry's law constant from H₂S solubility

((

H₂ S, a toxic gas with rotten egg like smell, is used for the qualitative analysis. If the solubility of H₂ S in water at STP is 0.195 m, calculate Henry's law constant.

Given

It is given that the solubility of H₂ S in water at STP is 0.195 m, i.e., 0.195 mol of H₂ S is dissolved in 1000 g of water.

$$= \frac{1000 \text{ g}}{18 \text{ g mol}^{-1}}$$

Moles of water

$$= 55.56 \text{ mol}$$

Key formula

$$= \frac{\text{Moles of H}_2\text{S}}{\text{Moles of H}_2\text{S} + \text{Moles of water}}$$



Substitute

$$= \frac{0.195}{0.195+55.56}$$

$$= 0.0035$$

Key formula

At STP, pressure (p) = 0.987 bar

According to Henry's law: p =



Conclusion

K_H x

$$\Rightarrow K_H = \frac{p}{x}$$

K_H = 282 bar //

$$= \frac{0.987}{0.0035} \text{ bar}$$

$$= 282 \text{ bar}$$

Q

7

Quantity of
CO₂
dissolved in
soda water



Henry's law constant for CO₂ in water is 1.67×10^8 Pa at 298 K. Calculate the quantity of CO₂ in 500 mL of soda water when packed under 2.5 atm CO₂ pressure at 298 K.

Given

It is given that:

$$K_H = 1.67 \times 10^8 \text{ Pa}$$

$$p_{\text{CO}_2} = 2.5 \text{ atm} = 2.5 \times 1.01325 \times 10^5 \text{ Pa}$$

$$= 2.533125 \times 10^5 \text{ Pa}$$

Key formula

According to Henry's law:

$$p_{\text{CO}_2} = K_H \times x$$

$$\Rightarrow x = \frac{p_{\text{CO}_2}}{K_H}$$

$$= \frac{2.533125 \times 10^5}{1.67 \times 10^8}$$

$$= 0.00152$$

Substitute

We can write,

n_{CO_2} is negligible as compared to $n_{\text{H}_2\text{O}}$ [Since,]

In 500 mL of soda water, the volume of water = 500 mL (the amount of soda present is neglected).

We can write:

500 mL of water = 500 g of water

$$= \frac{500}{18} \text{ mol of water}$$

$$= 27.78 \text{ mol of water}$$

$$\text{Now, } \frac{n_{\text{CO}_2}}{n_{\text{H}_2\text{O}}} = x$$

$$\frac{n_{\text{CO}_2}}{27.78} = 0.00152$$

$$n_{\text{CO}_2} = 0.042 \text{ mol}$$

Conclusion

Hence, quantity of CO₂ in 500 mL of soda water = $(0.042 \times 44) \text{ g}$

$$= 1.848 \text{ g}$$

$$\boxed{1.848 \text{ g of CO}_2} //$$

Q

8

Composition
of liquid and
vapour phase
from vapour
pressures



The vapour pressure of pure liquids A and B are 450 and 700 mm Hg respectively, at 350 K .

Find out the composition of the liquid mixture if total vapour pressure is 600 mm Hg .

Also find the composition of the vapour phase.

Given

It is given that:

$$p_A^0 = 450 \text{ mm of Hg}$$

$$p_B^0 = 700 \text{ mm of Hg}$$

$$p_{\text{total}} = 600 \text{ mm of Hg}$$

Key formula



From Raoult's law:

$$\Rightarrow p_{\text{total}} = p_A^0 x_A + p_B^0 (1 - x_A)$$

$$\Rightarrow p_{\text{total}} = p_A^0 x_A + p_B^0 - p_B^0 x_A$$

$$\Rightarrow p_{\text{total}} = (p_A^0 - p_B^0) x_A + p_B^0$$

$$\Rightarrow 600 = (450 - 700) x_A + 700$$

$$p_A = p_A^0 x_A$$

$$p_B = p_B^0 x_B = p_B^0 (1 - x_A)$$

Therefore, total

pressure,

$$p_{\text{total}} = p_A + p_B$$

Substitute

Therefore,

$$= 1 - 0.4$$

$$= 0.6$$

$$\text{Now, } p_A = p_A^0 x_A$$

$$= 450 \times 0.4 = 180 \text{ mm of Hg}$$

$$p_B = p_B^0 x_B$$

$$= 700 \times 0.6$$

$$= 420 \text{ mm of Hg. Now, in the vapour phase, mole fraction of liquid A} =$$

$$\frac{p_A}{p_A + p_B}$$

$$= \frac{180}{180 + 420}$$

$$= \frac{180}{600}$$

$$= 0.30$$

Conclusion

And, mole fraction of liquid B = 1 - 0.30

$$= 0.70$$

Liquid mixture: $x_A = 0.4$, $x_B = 0.6$; Vapour phase: mole fraction A = 0.30, mole fraction B = 0.70

Q

9

Vapour pressure lowering from urea solution



Vapour pressure of pure water at 298 K is 23.8 mm Hg. 50 g of urea (NH_2CONH_2) is dissolved in 850 g of water. Calculate the vapour pressure of water for this solution and its relative lowering.

Given

It is given that vapour pressure of water, $p_1^0 = 23.8$ mm of Hg

Weight of water taken, $w_1 = 850$ g

Weight of urea taken, $w_2 = 50$ g

Molecular weight of water, $M_1 = 18$ g mol⁻¹

Molecular weight of urea, $M_2 = 60$ g mol⁻¹

Now, we have to calculate vapour pressure of water in the solution. We take vapour pressure as p_1 .

Key formula

Now, from Raoult's law, we have:



$$\begin{aligned}\frac{p_1^0 - p_1}{p_1^0} &= \frac{n_2}{n_1 + n_2} \\ \Rightarrow \frac{p_1^0 - p_1}{p_1^0} &= \frac{w_2}{\frac{w_1}{M_1} + \frac{w_2}{M_2}} \\ \Rightarrow \frac{23.8 - p_1}{23.8} &= \frac{50}{\frac{850}{18} + \frac{50}{60}} \\ \Rightarrow \frac{23.8 - p_1}{23.8} &= \frac{0.83}{47.22 + 0.83} \\ \Rightarrow \frac{23.8 - p_1}{23.8} &= 0.0173 \\ \Rightarrow p_1 &= 23.4 \text{ mm of Hg}\end{aligned}$$

Conclusion

Hence, the vapour pressure of water in the given solution is 23.4 mm of Hg. Its relative lowering is 0.0173.

Vapour pressure = 23.4 mm of Hg; relative lowering = 0.0173

Q

10

Sucrose
needed to
raise boiling
point to 100°C

()

Boiling point of water at 750 mm Hg is 99.63°C.

How much sucrose is to be added to 500 g of water such that it boils at 100°C.

Given

Here, elevation of boiling point $\Delta T_b = (100+273)-(99.63+273)$

$$= 0.37 \text{ K}$$

Mass of water, $w_1 = 500 \text{ g}$

Molar mass of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$), $M_2 = 11 \times 12 + 22 \times 1 + 11 \times 16$

$$= 342 \text{ g mol}^{-1}$$

Molal elevation constant, $K_b = 0.52 \text{ K kg mol}^{-1}$. We know that:

Key formula

$$\Delta T_b = \frac{K_b \times 1000 \times w_2}{M_2 \times w_1}$$



$$\Rightarrow w_2 = \frac{\Delta T_b \times M_2 \times w_1}{K_b \times 1000}$$

$$= \frac{0.37 \times 342 \times 500}{0.52 \times 1000}$$

$$= 121.67 \text{ g (approximately)}$$

Conclusion

Hence, 121.67 g of sucrose is to be added.

Note: There is a slight variation in this answer and the one given in the NCERT textbook.

121.67 g of sucrose //

Q

11

Mass of
ascorbic acid
for a given
freezing
point
depression



Calculate the mass of ascorbic acid (Vitamin C, $\text{C}_6\text{H}_8\text{O}_6$) to be dissolved in 75 g of acetic acid to lower its melting point by 1.5°C. $K_f = 3.9 \text{ K kg mol}^{-1}$.

Given

Mass of acetic acid, $w_1 = 75 \text{ g}$

Molar mass of ascorbic acid ($\text{C}_6\text{H}_8\text{O}_6$), $M_2 = 6 \times 12 + 8 \times 1 + 6 \times 16$

$$= 176 \text{ g mol}^{-1}$$

Lowering of melting point, $\Delta T_f = 1.5 \text{ K}$. We know that:

Key formula

$$\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$$



$$\Rightarrow w_2 = \frac{\Delta T_f \times M_2 \times w_1}{K_f \times 1000}$$

$$= \frac{1.5 \times 176 \times 75}{3.9 \times 1000}$$

$$= 5.08 \text{ g (approx)}$$

Conclusion

Hence, 5.08 g of ascorbic acid is needed to be dissolved.

Note: There is a slight variation in this answer and the one given in the NCERT textbook.

5.08 g of ascorbic acid //

Q

12

Osmotic
pressure of a
polymer
solution



Calculate the osmotic pressure in pascals exerted by a solution prepared by dissolving 1.0 g of polymer of molar mass 185,000 in 450 mL of water at 37°C.

Given

It is given that:

Volume of water, $V = 450 \text{ mL} = 0.45 \text{ L}$

Temperature, $T = (37+273) \text{ K} = 310 \text{ K}$

Number of moles of the polymer, $n = \frac{1}{185000} \text{ mol}$

We know that:

Key formula

Osmotic pressure,



$$\pi = \frac{n}{V} R T$$

Conclusion

$$= \frac{1}{185000} \text{ mol} \times \frac{1}{0.45 \text{ L}} \times 8.314 \times 10^3 \text{ PaL K}^{-1} \text{ mol}^{-1} \times 310 \text{ K}$$

$$= 30.98 \text{ Pa}$$

$$= 31 \text{ Pa (approximately)}$$

$$\pi = 31 \text{ Pa (approximately)} //$$

Part 3 • Questions and Solutions

Q

1

Definition
and types of
solution



Define the term solution. How many types of solutions are formed? Write briefly about each type with an example.

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

A solution is a homogeneous mixture of two or more chemically non-reacting substances, in which one substance (the solute) is dissolved in another (the solvent); its composition and properties are uniform throughout.

Solutions are classified into three broad types based on the physical state of the solvent, and each of these has three further types based on the physical state of the solute:

Gaseous solutions (solvent is a gas) -- e.g. a mixture of oxygen and nitrogen gases (gas in gas); chloroform vapour mixed with nitrogen gas (liquid in gas); camphor vapour in nitrogen gas (solid in gas).

Liquid solutions (solvent is a liquid) -- e.g. oxygen dissolved in water (gas in liquid); ethanol dissolved in water (liquid in liquid); glucose dissolved in water (solid in liquid).

Solid solutions (solvent is a solid) -- e.g. hydrogen dissolved in palladium (gas in solid); mercury in gold, an amalgam (liquid in solid); copper dissolved in gold, an alloy (solid in solid).

A solution is a homogeneous mixture of two or more substances; solutions are of three broad types -- gaseous, liquid and solid -- each further divided by the physical state of the solute, e.g. oxygen in water (gas in liquid), ethanol in water (liquid in liquid), glucose in water (solid in liquid).

Q

2

Solid
solution with
a gaseous
solute

(())

Give an example of a solid solution in which the solute is a gas.

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

A solid solution in which the solute is a gas is formed, for example, when hydrogen gas is absorbed (dissolved) in palladium metal -- written as H_2 in Pd.

Hydrogen gas dissolved in palladium metal (H_2 in Pd) is a solid solution with a gaseous solute.

Q

3

Definitions:
mole
fraction,
molality,
molarity,
mass
percentage



Define the following terms:

Part (i)

Mole fraction

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

Mole fraction of a component in a solution is the ratio of the number of moles of that component to the total number of moles of all components present. For a two-component solution with n_1 mol of solvent and n_2 mol of solute,

$$x_1 = \frac{n_1}{n_1 + n_2}, \quad x_2 = \frac{n_2}{n_1 + n_2}$$

Mole fraction does not depend on temperature.

Mole fraction $x =$

$$\frac{\text{moles of the component}}{\text{total moles of all components}}$$

Part (ii)

Molality

Conclusion

Molality (m) is the number of moles of solute dissolved per kilogram of solvent:

$$\text{Molality} = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

Since it is defined using the mass (not volume) of solvent, molality does not change with temperature.

Molality =

$$\frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$$

, unit mol kg⁻¹.

Part (iii)

Molarity

Conclusion

Molarity (M) is the number of moles of solute dissolved per litre of solution:

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution in litres}}$$

Because the volume of a solution changes slightly with temperature, molarity is temperature-dependent.

Molarity =

$$\frac{\text{moles of solute}}{\text{volume of solution (L)}}$$

, unit mol L⁻¹.

Part (iv)

Mass percentage.

Conclusion

Mass percentage of a component is the mass of that component per 100 g of solution:

$$\text{Mass \%} = \frac{\text{mass of the component}}{\text{total mass of the solution}} \times 100\%$$

Mass % =

$$\frac{\text{mass of the component}}{\text{total mass of solution}} \times 100.$$

Q

4

Molarity of concentrated nitric acid



Concentrated nitric acid used in laboratory work is 68% nitric acid by mass in aqueous solution. What should be the molarity of such a sample of the acid if the density of the solution is 1.504 g mL^{-1} ?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Given

Consider 1 litre (1000 mL) of the acid solution. Its mass is $1000 \text{ mL} \times 1.504 \text{ g mL}^{-1} = 1504 \text{ g}$. Mass of HNO_3 present (68% by mass): $0.68 \times 1504 \text{ g} = 1022.72 \text{ g}$. Molar mass of $\text{HNO}_3 = 1+14+3 \times 16 = 63 \text{ g mol}^{-1}$.

Substitute

$$\text{Moles of } \text{HNO}_3 = \frac{1022.72}{63} = 16.23 \text{ mol}$$

This is the amount of acid present in 1 litre of solution.

Conclusion

$$\text{Molarity} = \frac{16.23 \text{ mol}}{1 \text{ L}} = 16.23 \text{ M}$$

16.23 M //

Q

5

Molality, mole fraction and molarity of glucose solution



A solution of glucose in water is labelled as 10% w/w, what would be the molality and mole fraction of each component in the solution? If the density of solution is 1.2 g mL^{-1} , then what shall be the molarity of the solution?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Given

A 10% w/w glucose solution means 10 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$, $M=180 \text{ g mol}^{-1}$) is present in 100 g of solution, i.e. in $(100-10)=90 \text{ g}$ of water.

Substitute

$$\text{Moles of glucose} = \frac{10}{180} = 0.0556 \text{ mol}; \text{ mass of water} = 0.090 \text{ kg}.$$

$$\text{Molality} = \frac{0.0556 \text{ mol}}{0.090 \text{ kg}} = 0.617 \text{ mol kg}^{-1}$$

$$\text{Moles of water} = \frac{90}{18} = 5 \text{ mol}.$$

$$x_{\{\text{glucose}\}} = \frac{0.0556}{0.0556+5} = 0.0111, \quad x_{\{\text{water}\}} = 1-0.0111 = 0.9889$$

Volume of 100 g of solution (density 1.2 g mL^{-1}):

$$\frac{100 \text{ g}}{1.2 \text{ g mL}^{-1}} = 83.33 \text{ mL} = 0.08333 \text{ L}$$

Conclusion

$$\text{Molarity} = \frac{0.0556 \text{ mol}}{0.08333 \text{ L}} = 0.667 \text{ M}$$

Molality = $0.617 \text{ mol kg}^{-1}$; mole fraction of glucose = 0.0111 (water = 0.9889); molarity = 0.667 M //

Q

6

Volume of
HCl to react
with a
carbonate/bi
carbonate
mixture

()

How many mL of 0.1 M HCl are required to react completely with 1 g mixture of Na_2CO_3 and NaHCO_3 containing equimolar amounts of both?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Given

Let n mol each of Na_2CO_3 ($M=106 \text{ g mol}^{-1}$) and NaHCO_3 ($M=84 \text{ g mol}^{-1}$) be present (equimolar mixture, total mass 1 g):

$$106n + 84n = 1 \Rightarrow 190n = 1 \Rightarrow n = 5.263 \times 10^{-3} \text{ mol}$$

Both salts react with HCl: $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ and $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$.

Substitute

Na_2CO_3 needs $2n$ mol HCl and NaHCO_3 needs n mol HCl -- $3n$ mol in total:

$$3 \times 5.263 \times 10^{-3} = 1.579 \times 10^{-2} \text{ mol HCl}$$

Conclusion

$$\frac{1.579 \times 10^{-2} \text{ mol}}{0.1 \text{ mol L}^{-1}} = 0.1579 \text{ L} = 157.9 \text{ mL}$$

157.9 mL

///

Q

7

Mass
percentage
of a mixed
solution



A solution is obtained by mixing 300 g of 25% solution and 400 g of 40% solution by mass.

Calculate the mass percentage of the resulting solution.

Given

Total amount of solute present in the mixture is given by,

$$\begin{aligned} 300 \times \frac{25}{100} + 400 \times \frac{40}{100} \\ = 75 + 160 \\ = 235 \text{ g} \end{aligned}$$

Substitute

Total amount of solution = $300 + 400 = 700 \text{ g}$

$$\begin{aligned} \text{Therefore, mass percentage (w / w) of the solute in the resulting solution,} &= \frac{235}{700} \\ &\times 100 \% \\ &= 33.57\% \end{aligned}$$

Conclusion

$$\begin{aligned} \text{And, mass percentage (w / w) of the solvent in the resulting solution,} \\ = (100 - 33.57) \% \\ = 66.43 \% \end{aligned}$$

Q

8

Molality and
molarity of
an antifreeze
solution



An antifreeze solution is prepared from 222.6 g of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) and 200 g of water. Calculate the molality of the solution. If the density of the solution is 1.072 g mL^{-1} , then what shall be the molarity of the solution?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Given

Molar mass of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) = 62 g mol^{-1} .

$$\text{Moles of } \text{C}_2\text{H}_6\text{O}_2 = \frac{222.6}{62} = 3.590 \text{ mol}$$

Mass of water = 200 g = 0.200 kg.

Substitute

$$\text{Molality} = \frac{3.590 \text{ mol}}{0.200 \text{ kg}} = 17.95 \text{ mol kg}^{-1}$$

Total mass of solution = 222.6 + 200 = 422.6 g; with density 1.072 g mL⁻¹:

$$\text{Volume} = \frac{422.6 \text{ g}}{1.072 \text{ g mL}^{-1}} = 394.2 \text{ mL} = 0.3942 \text{ L}$$

Conclusion

$$\text{Molarity} = \frac{3.590 \text{ mol}}{0.3942 \text{ L}} = 9.11 \text{ M}$$

Molality = 17.95 mol kg⁻¹; molarity = 9.11 M

//

Q

9

Chloroform contamination: percent by mass and molality



A sample of drinking water was found to be severely contaminated with chloroform (CHCl₃) supposed to be a carcinogen. The level of contamination was 15 ppm (by mass):

Part (i)

express this in percent by mass

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

A contamination level of 15 ppm by mass means 15 g of CHCl₃ per 10⁶ g of water, i.e.

$$\frac{15}{10^6} \times 100\% = 0.0015\%$$

0.0015% by mass

//

Part (ii)

determine the molality of chloroform in the water sample.

Conclusion

Molar mass of CHCl₃ = 12 + 1 + 3 × 35.5 = 119.5 g mol⁻¹. In 10⁶ g of (essentially) water, the 15 g of chloroform is

$$\frac{15}{119.5} = 0.1255 \text{ mol}$$

and the mass of water is effectively 10⁶ g = 1000 kg (the dissolved chloroform's own mass is negligible next to it), so

$$\text{Molality} = \frac{0.1255 \text{ mol}}{1000 \text{ kg}} = 1.25 \times 10^{-4} \text{ mol kg}^{-1}$$

1.25 × 10⁻⁴ mol kg⁻¹

//

Q

10

Molecular interaction in alcohol-water solution

(())

What role does the molecular interaction play in a solution of alcohol and water?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

In pure ethanol and in pure water, each liquid's own molecules are held together by strong hydrogen bonding. When they are mixed, ethanol's -OH group can still hydrogen-bond with water, but these alcohol-water interactions are somewhat weaker than the original alcohol-alcohol and water-water hydrogen bonds. This makes it easier for molecules to escape into the vapour phase, so the solution's vapour pressure is higher than

The alcohol-water hydrogen bonds are weaker than the pure-liquid hydrogen bonds they replace, so the solution shows positive deviation from Raoult's law.

Raoult's law predicts, and an ethanol-water solution shows positive deviation from ideal behaviour.

Q

11

Gas solubility decreasing with temperature



Why do gases always tend to be less soluble in liquids as the temperature is raised?

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

Dissolution of a gas in a liquid is generally an exothermic process ($\Delta H < 0$) -- it is accompanied by the release of heat, similar in spirit to condensation. By Le Chatelier's principle, raising the temperature shifts this dynamic equilibrium in the direction that absorbs the added heat, i.e. towards the gas coming back out of solution. The dissolved gas molecules also gain kinetic energy at higher temperature, making it easier for them to escape the liquid. Both effects make gases less soluble in liquids as the temperature is raised.

Gas dissolution in a liquid is exothermic, so by Le Chatelier's principle its solubility falls as temperature rises.

Q

12

Henry's law statement and applications



State Henry's law and mention some important applications.

Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Conclusion

Henry's law states that, at a constant temperature, the partial pressure of a gas in the vapour phase is directly proportional to the mole fraction of the gas dissolved in the solution:

$$p = K_H \cdot x$$

where K_H is the Henry's law constant for that gas-solvent pair.

Some important applications:

Carbonated drinks are bottled under high CO_2 pressure to keep more gas dissolved; the fizz appears when the bottle is opened and the pressure drops.

Scuba divers' breathing tanks are diluted with helium (rather than nitrogen), because nitrogen's higher solubility under the increased pressure at depth can cause the 'bends' when it comes out of solution too quickly on ascent.

At high altitude, the lower partial pressure of oxygen means less oxygen dissolves in the blood, which can cause altitude sickness (anoxia) in climbers and people who travel to high altitudes.

$p = K_H \cdot x$; applications include bottling carbonated drinks, scuba divers' gas tanks, and altitude sickness from low oxygen partial pressure.

Q

13

Partial
pressure of
ethane from
Henry's law



The partial pressure of ethane over a solution containing 6.56×10^{-3} g of ethane is 1 bar. If the solution contains 5.00×10^{-2} g of ethane, then what shall be the partial pressure of the gas?

Given

Molar mass of ethane (C_2H_6) = $2 \times 12 + 6 \times 1$

$$= 30 \text{ g mol}^{-1}$$

$\therefore$ Number of moles present in 6.56×10^{-2} g of ethane

$$= \frac{6.56 \times 10^{-2}}{30}$$

$$= 2.187 \times 10^{-3} \text{ mol}$$

Key formula

Let the number of moles of the solvent be x. By Henry's law,

$$p = K_H x$$

$$\Rightarrow 1 \text{ bar} = K_H \cdot \frac{2.187 \times 10^{-3}}{2.187 \times 10^{-3} + x}$$

$$\Rightarrow 1 \text{ bar} = K_H \frac{2.187 \times 10^{-3}}{x}$$

$$\Rightarrow K_H = \frac{x}{2.187 \times 10^{-3}} \text{ bar (Since } x \gg 2.187 \times 10^{-3})$$

Substitute

Number of moles present in 5.00×10^{-2} g of ethane

$$= \frac{5.00 \times 10^{-2}}{30} \text{ mol}$$

$$= 1.67 \times 10^{-3} \text{ mol.}$$

Applying Henry's law again,

$$p = K_H x$$

$$= \frac{x}{2.187 \times 10^{-3}} \times \frac{1.67 \times 10^{-3}}{(1.67 \times 10^{-3}) + x}$$

$$= \frac{x}{2.187 \times 10^{-3}} \times \frac{1.67 \times 10^{-3}}{x} \text{ (Since, } x \gg 1.67 \times 10^{-3})$$

$$= 0.764 \text{ bar}$$

Conclusion

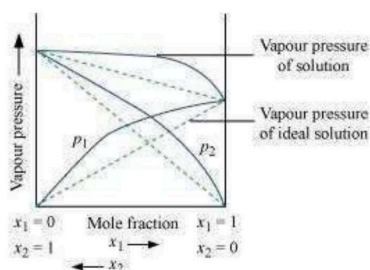
So the partial pressure of the gas is 0.764 bar. **0.764 bar** //

Q

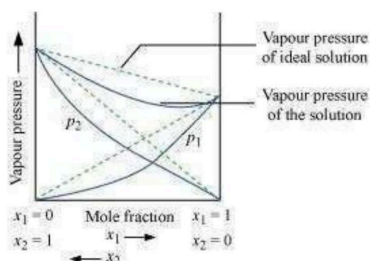
14

Positive / negative deviation from Raoult's law and mixing enthalpy

()



Vapour pressure of a two-component solution showing positive deviation from Raoult's law



Vapour pressure of a two-component solution showing negative deviation from Raoult's law

What is meant by positive and negative deviations from Raoult's law and how is the sign of $\Delta_{\text{mix}} H$ related to positive and negative deviations from Raoult's law?

Given

By Raoult's law, each volatile component's partial vapour pressure in a solution is directly proportional to its mole fraction.

A solution that obeys Raoult's law at every concentration is called an ideal solution.

A solution that does not obey Raoult's law is non-ideal, and its vapour pressure is either higher or lower than Raoult's law predicts.

If the vapour pressure is higher, the solution shows positive deviation.

If it is lower, the solution shows negative deviation.

For an ideal solution, the enthalpy of mixing the pure components is zero.

$$\Delta_{\text{mix}} H = 0$$

For a solution with positive deviation, heat is absorbed on mixing.

$$\therefore \Delta_{\text{mix}} H = \text{Positive}$$

For a solution with negative deviation, heat is released on mixing.

$$\therefore \Delta_{\text{mix}} H = \text{Negative}$$

Q

15

Molar mass of solute from vapour pressure



An aqueous solution of 2% non-volatile solute exerts a pressure of 1.004 bar at the normal boiling point of the solvent.

What is the molar mass of the solute?

Given

Here,

Vapour pressure of the solution at normal boiling point (p_1) = 1.004 bar

Vapour pressure of pure water at normal boiling point (p_1^0) = 1.013 bar

Mass of solute, (w_2) = 2 g

Mass of solvent (water), (w_1) = 98 g

Molar mass of solvent (water), (M_1) = 18 g mol^{-1} .

Substitute

By Raoult's law,

$$\begin{aligned}\frac{p_1^0 - p_1}{p_1^0} &= \frac{w_2 \times M_1}{M_2 \times w_1} \\ \Rightarrow \frac{1.013 - 1.004}{1.013} &= \frac{2 \times 18}{M_2 \times 98} \\ \Rightarrow \frac{0.009}{1.013} &= \frac{2 \times 18}{M_2 \times 98} \\ \Rightarrow M_2 &= \frac{1.013 \times 2 \times 18}{0.009 \times 98} \\ &= 41.35 \text{ g mol}^{-1}\end{aligned}$$

Conclusion

So the molar mass of the solute is 41.35 g mol^{-1} . **41.35 g mol^{-1}** //

Q

16

Vapour pressure of heptane-octane ideal solution



Heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components are 105.2 kPa and 46.8 kPa respectively. What will be the vapour pressure of a mixture of 26.0 g of heptane and 35 g of octane?

Given

Vapour pressure of

(p_1^0) = 105.2 kPa

heptane heptane

Vapour pressure of (p_2^0) = 46.8 kPa

octane

Substitute

We know that,

Molar mass of heptane (C_7H_{16}) = $7 \times 12 + 16 \times 1$

= 100 g mol^{-1}

$\therefore$ Number of moles of heptane = $\frac{26}{100} \text{ mol}$

= 0.26 mol

Molar mass of octane (C_8H_{18}) = $8 \times 12 + 18 \times 1$

= 114 g mol^{-1}

$\therefore$ Number of moles of octane = $\frac{35}{114} \text{ mol}$

= 0.31 mol

Mole fraction of heptane, x_1 = $\frac{0.26}{0.26 + 0.31}$

= 0.456

And, mole fraction of octane, x_2 = $1 - 0.456$

= 0.544

Now, partial pressure of heptane, $p_1 = x_1 p_1^0$

$$= 0.456 \times 105.2$$

$$= 47.97 \text{ kPa}$$

$$\text{Partial pressure of octane, } p_2 = x_2 p_2^0$$

$$= 0.544 \times 46.8$$

$$= 25.46 \text{ kPa}$$

Conclusion

$$\text{Hence, vapour pressure of solution, } p_{\text{total}} = p_1 + p_2$$

$$= 47.97 + 25.46$$

$$= 73.43 \text{ kPa} //$$

$$= 73.43 \text{ kPa}$$

Q

17

Vapour pressure of a 1 molal non-volatile solute solution



The vapour pressure of water is 12.3 kPa at 300 K .

Calculate vapour pressure of 1 molal solution of a non-volatile solute in it.

Given

1 molal solution means 1 mol of the solute is present in 1000 g of the solvent (water).

$$\text{Molar mass of water} = 18 \text{ g mol}^{-1}$$

Substitute

$$\therefore \text{Number of moles present in 1000 g of water} = \frac{1000}{18}$$

$$= 55.56 \text{ mol}$$

Therefore, mole fraction of the solute in the solution is

$$x_2 = \frac{1}{1+55.56} = 0.0177 .$$

Given

It is given that,

$$\text{Vapour pressure of water, } p_1^0 = 12.3 \text{ kPa}$$

Key formula

Applying the relation,

$$\frac{p_1^0 - p_1}{p_1^0} = x_2$$

Substitute

$$\Rightarrow \frac{12.3 - p_1}{12.3} = 0.0177$$

$$\Rightarrow p_1 = 12.0823$$

$$= 12.08 \text{ kPa (approximately)}$$

$$\Rightarrow 12.3 - p_1 = 0.2177$$

Conclusion

So the vapour pressure of the solution is 12.08 kPa.

$$= 12.08 \text{ kPa} //$$

Q

18

Mass of solute to reduce vapour pressure by 20%

()

Calculate the mass of a non-volatile solute (molar mass 40 g mol^{-1}) which should be dissolved in 114 g octane to reduce its vapour pressure to 80 %.

Given

Let the vapour pressure of pure octane be p_1^0 .

So the vapour pressure of octane after adding the solute is

$$\frac{80}{100} p_1^0 = 0.8 p_1^0$$

Molar mass of solute, $M_2 = 40 \text{ g mol}^{-1}$

Mass of octane, $w_1 = 114 \text{ g}$

Molar mass of octane, ($\text{C}_8 \text{H}_{18}$), $M_1 = 8 \times 12 + 18 \times 1$

$$= 114 \text{ g mol}^{-1}$$

Substitute

Applying the relation,

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{w_2 \times M_1}{M_2 \times w_1}$$

$$\Rightarrow \frac{p_1^0 - 0.8 p_1^0}{p_1^0} = \frac{w_2 \times 114}{40 \times 114}$$

$$\Rightarrow \frac{0.2 p_1^0}{p_1^0} = \frac{w_2}{40}$$

$$\Rightarrow 0.2 = \frac{w_2}{40}$$

$$\Rightarrow w_2 = 8 \text{ g}$$

Conclusion

So the required mass of the solute is 8 g. **8 g** //

Q

19

Molar mass of solute and vapour pressure of water



A solution containing 30 g of non-volatile solute exactly in 90 g of water has a vapour pressure of 2.8 kPa at 298 K.

Further, 18 g of water is then added to the solution and the new vapour pressure becomes 2.9 kPa at 298 K.

Calculate:

Part (i)

molar mass of the solute

Given

(i) Let, the molar mass of the solute be $M \text{ g mol}^{-1}$

$$n_1 = \frac{90 \text{ g}}{18 \text{ g mol}^{-1}} = 5 \text{ mol}$$

Now, the no. of moles of solvent (water),

$$n_2 = \frac{30 \text{ g}}{M \text{ mol}^{-1}} = \frac{30}{M} \text{ mol}$$

And, the no. of moles of solute,

$$p_1 = 2.8 \text{ kPa}$$

Applying the relation:

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2}$$

$$\Rightarrow \frac{p_1^0 - 2.8}{p_1^0} = \frac{30}{M}$$

$$\Rightarrow 1 - \frac{2.8}{p_1^0} = \frac{30}{5M + 30}$$

$$\Rightarrow 1 - \frac{2.8}{p_1^0} = \frac{30}{5M + 30}$$

$$\Rightarrow \frac{2.8}{p_1^0} = 1 - \frac{30}{5M + 30}$$

$$\Rightarrow \frac{2.8}{p_1^0} = \frac{5M + 30 - 30}{5M + 30}$$

$$\Rightarrow \frac{2.8}{p_1^0} = \frac{5M}{5M + 30}$$

$$\Rightarrow \frac{p_1^0}{2.8} = \frac{5M + 30}{5M}$$

Substitute

After the addition of 18 g of water:

$$n_1 = \frac{90 + 18 \text{ g}}{18} = 6 \text{ mol}$$

$$p_1 = 2.9 \text{ kPa}$$

Again, applying the relation:

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_2}{n_1 + n_2}$$

$$\Rightarrow \frac{p_1^0 - 2.9}{p_1^0} = \frac{30}{M}$$

$$\Rightarrow 1 - \frac{2.9}{p_1^0} = \frac{30}{6M + 30}$$

$$\Rightarrow 1 - \frac{2.9}{p_1^0} = \frac{30}{6M + 30}$$

$$\Rightarrow \frac{2.9}{p_1^0} = 1 - \frac{30}{6M + 30}$$

$$\Rightarrow \frac{2.9}{p_1^0} = \frac{6M+30-30}{6M+30}$$

$$\Rightarrow \frac{2.9}{p_1^0} = \frac{6M}{6M+30}$$

$$\Rightarrow \frac{p_1^0}{2.9} = \frac{6M+30}{6M}$$

Dividing equation (i) by (ii), we have:

$$\frac{2.9}{2.8} = \frac{\frac{5M+30}{5M}}{\frac{6M+30}{6M}}$$

$$\Rightarrow \frac{2.9}{2.8} \times \frac{6M+30}{6} = \frac{5M+30}{5}$$

$$\Rightarrow 2.9 \times 5 \times (6M+30) = 2.8 \times 6 \times (5M+30)$$

$$\Rightarrow 87M+435 = 84M+504$$

$$\Rightarrow 3M = 69$$

$$\Rightarrow M = 23 \text{ u}$$

Conclusion

Therefore, the molar mass of the solute is 23 g mol^{-1} . **23 g mol⁻¹** //

Part (ii)

vapour pressure of water at 298 K.

Substitute

(ii) Putting the value of 'M' in equation (i), we have:

$$\frac{p_1^0}{2.8} = \frac{5 \times 23+30}{5 \times 23}$$

$$\Rightarrow \frac{p_1^0}{2.8} = \frac{145}{115}$$

$$\Rightarrow p_1^0 = 3.53$$

Conclusion

Hence, the vapour pressure of water at 298 K is 3.53 kPa. **3.53 kPa** //

Q

20

Freezing point of glucose solution from cane sugar data



A 5% solution (by mass) of cane sugar in water has freezing point of 271 K. Calculate the freezing point of 5% glucose in water if freezing point of pure water is 273.15 K.

Given

Here, $\Delta T_f = (273.15-271) \text{ K}$

$$= 2.15 \text{ K}$$

Molar mass of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) = $12 \times 12 + 22 \times 1 + 11 \times 16$

$$= 342 \text{ g mol}^{-1}$$

5% solution (by mass) of cane sugar in water means 5 g of cane sugar is present in (100-5) g = 95 g of water.

Substitute

$$\text{Now, number of moles of cane sugar} = \frac{5}{342} \text{ mol}$$

$$= 0.0146 \text{ mol}$$

Therefore, molality of the solution,

$$m = \frac{0.0146 \text{ mol}}{0.095 \text{ kg}}$$

$$= 0.1537 \text{ mol kg}^{-1}$$

Key formula



Applying the relation,

$$\Delta T_f = K_f \times m$$

$$\Rightarrow K_f = \frac{\Delta T_f}{m}$$

$$= \frac{2.15 \text{ K}}{0.1537 \text{ mol kg}^{-1}}$$

$$= 13.99 \text{ K kg mol}^{-1}$$

Given



Molar mass of glucose ($\text{C}_6 \text{H}_{12} \text{O}_6$) = $6 \times 12 + 12 \times 1 + 6 \times 16 =$

$$180 \text{ g mol}^{-1}$$

5% glucose in water means 5 g of glucose is present in (100-5) g = 95 g of water.

Substitute

$$\therefore \text{Number of moles of glucose} = \frac{5}{180} \text{ mol}$$

$$= 0.0278 \text{ mol}$$

Therefore, molality of the solution,

$$m = \frac{0.0278 \text{ mol}}{0.095 \text{ kg}}$$

$$= 0.2926 \text{ mol kg}^{-1}$$

Applying the relation,

$$\Delta T_f = K_f \times m$$

$$= 13.99 \text{ K kg mol}^{-1} \times 0.2926 \text{ mol kg}^{-1}$$

$$= 4.09 \text{ K (approximately)}$$

Conclusion

So the freezing point of the 5% glucose solution is $(273.15 - 4.09) \text{ K} = 269.06 \text{ K}$.

269.06 K //

Q

21

Atomic masses of A and B from freezing point depression



Two elements A and B form compounds having formula AB_2 and AB_4 . When dissolved in 20 g of benzene (C_6H_6), 1 g of AB_2 lowers the freezing point by 2.3 K whereas 1.0 g of AB_4 lowers it by 1.3 K. The molar depression constant for benzene is $5.1 \text{ K kg mol}^{-1}$. Calculate atomic masses of A and B.

Substitute

First, we find the molar mass of AB_2 and AB_4 using the freezing point depression formula:

$$M_2 = \frac{1000 \times w_2 \times k_f}{\Delta T_f \times w_1}$$

$$M_{AB_2} = \frac{1000 \times 1 \times 5.1}{2.3 \times 20}$$

$$= 110.87 \text{ g mol}^{-1}$$

$$M_{AB_4} = \frac{1000 \times 1 \times 5.1}{1.3 \times 20}$$

$$= 196.15 \text{ g mol}^{-1}$$

Now, we have the molar masses of AB_2 and AB_4 as $110.87 \text{ g mol}^{-1}$ and $196.15 \text{ g mol}^{-1}$ respectively.

Let the atomic masses of A and B be x and y respectively.

Now, we can write:

$$x + 2y = 110.87$$

$$x + 4y = 196.15$$

Subtracting equation (i) from (ii), we have

$$2y = 85.28$$

$$\Rightarrow y = 42.64$$

Conclusion

Putting the value of 'y' in equation (1), we have x

$$+ 2 \times 42.64 = 110.87$$

$$\Rightarrow x = 25.59$$

Hence, the atomic masses of A and B are 25.59 u and 42.64 u respectively.



Q

22

Concentration of glucose solution from osmotic pressure

(())

At 300 K, 36 g of glucose present in a litre of its solution has an osmotic pressure of 4.98 bar.

If the osmotic pressure of the solution is 1.52 bars at the same temperature, what would be its concentration?

Given

We are given:

$$T = 300 \text{ K}$$

$$\Pi = 1.52 \text{ bar}$$

$$R = 0.083 \text{ bar L K}^{-1} \text{ mol}^{-1}$$

Key formulaApplying the relation, $\Pi =$ 

CRT

Substitute

$$\Rightarrow C = \frac{\pi}{R T}$$

$$= \frac{1.52 \text{ bar}}{0.083 \text{ bar L K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}}$$

$$= 0.061 \text{ mol}$$

Since the volume of the solution is 1 L, the concentration of the solution would be 0.061 M.

//

Q**23**

Dominant
intermolecular
or
interaction
in given
pairs



Suggest the most important type of intermolecular attractive interaction in the following pairs.

Part (i)

n-hexane and n-octane**Conclusion**

Van der Waals' forces of attraction.

Part (ii)

I₂ and CCl₄**Conclusion**

Van der Waals' forces of attraction.

Part (iii)

NaClO₄ and water**Conclusion**

Ion-dipole interaction.

Part (iv)

methanol and acetone**Conclusion**

Dipole-dipole interaction.

Part (v)

acetonitrile (CH₃ CN) and acetone (C₃ H₆ O).**Conclusion**

Dipole-dipole interaction.

Q**24**

Increasing
solubility
order in n-
octane



Based on solute-solvent interactions, arrange the following in order of increasing solubility in n-octane and explain. Cyclohexane, KCl, CH₃ OH, CH₃ CN.

Conclusion

n-octane is a non-polar solvent, so it dissolves non-polar solutes more easily than polar solutes in the n-octane.

The order of increasing polarity is:

Cyclohexane < CH₃ CN < CH₃ OH < KCl

Therefore, the order of increasing solubility is: KCl < CH₃ OH < CH₃ CN < Cyclohexane

//

Q

25

Solubility
classification
of given
compounds
in water



Amongst the following compounds, identify which are insoluble, partially soluble and highly soluble in water?

Part (i)

phenol

Conclusion

Phenol (C_6H_5OH) has the polar group $-OH$ and non-polar group $-C_6H_5$. Thus, phenol is partially soluble in water.

Part (ii)

toluene

Conclusion

Toluene ($C_6H_5-CH_3$) has no polar groups. Thus, toluene is insoluble in water.

Part (iii)

formic acid

Conclusion

Formic acid ($HCOOH$) has the polar group $-OH$ and can form H-bond with water. Thus, formic acid is highly soluble in water.

Part (iv)

ethylene glycol

Conclusion

Ethylene glycol ($HO-CH_2-CH_2-OH$) has polar $-OH$ group and can form H-bond. Thus, it is highly soluble in water.

Part (v)

chloroform

Conclusion

Chloroform is insoluble in water.

Part (vi)

pentanol.

Conclusion

Pentanol ($C_5H_{11}OH$) has polar $-OH$ group, but it also contains a very bulky nonpolar $-C_5H_{11}$ group. Thus, pentanol is partially soluble in water.

Q

26

Molarity of
 Na^+ ions in
lake water

()

If the density of some lake water is 1.25 g mL^{-1} and contains 92 g of Na^+ ions per kg of water, calculate the molarity of Na^+ ions in the lake.

Conclusion

$$\begin{aligned} \text{Number of moles present in 92 g of } Na^+ \text{ ions} &= \frac{92 \text{ g}}{23 \text{ g mol}^{-1}} \\ &= 4 \text{ mol} \\ &= \frac{4 \text{ mol}}{1 \text{ kg}} \\ &= 4 \text{ m} \end{aligned}$$

Q

27

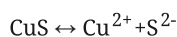
Maximum molarity of CuS from solubility product



If the solubility product of CuS is 6×10^{-16} , calculate the maximum molarity of CuS in aqueous solution.

Given

Solubility product of CuS, $K_{sp} = 6 \times 10^{-16}$. Let s be the solubility of CuS in mol L^{-1} .



S S

Key formula

Now,

$$K_{sp} = [\text{Cu}^{2+}][\text{S}^{2-}]$$

Substitute

$$= s \times s$$

$$= s^2$$

$$\text{So, } K_{sp} = s^2 = 6 \times 10^{-16}$$

$$\Rightarrow s = \sqrt{6 \times 10^{-16}}$$

Conclusion

$$= 2.45 \times 10^{-8} \text{ mol L}^{-1}$$

Hence, the maximum molarity of CuS in an aqueous solution is $2.45 \times 10^{-8} \text{ mol L}^{-1}$.

//

Q

28

Mass percentage of aspirin in acetonitrile



Calculate the mass percentage of aspirin ($\text{C}_9\text{H}_8\text{O}_4$) in acetonitrile (CH_3CN) when 6.5 g of $\text{C}_9\text{H}_8\text{O}_4$ is dissolved in 450 g of CH_3CN .

Conclusion

6.5 g of $\text{C}_9\text{H}_8\text{O}_4$ is dissolved in 450 g of CH_3CN .

Then, total mass of the solution = $(6.5 + 450) \text{ g}$

$$= 456.5 \text{ g}$$

$$\text{Therefore, mass percentage of } \text{C}_9\text{H}_8\text{O}_4 = \frac{6.5}{456.5} \times 100\% = 1.424\%$$

//

Q

29

Mass of nalorphene solution for a given dose



Nalorphene ($\text{C}_{19}\text{H}_{21}\text{NO}_3$), similar to morphine, is used to combat withdrawal symptoms in narcotic users. Dose of nalorphene generally given is 1.5 mg. Calculate the mass of $1.5 \times 10^{-3} \text{ m}$ aqueous solution required for the above dose.

Given

The molar mass of nalorphene ($\text{C}_{19}\text{H}_{21}\text{NO}_3$) is given as:

$$19 \times 12 + 21 \times 1 + 1 \times 14 + 3 \times 16 = 311 \text{ g mol}^{-1}$$

Substitute

In $1.5 \times 10^{-3} \text{ m}$ aqueous solution of nalorphene,

$$1 \text{ kg (1000 g) of water contains } 1.5 \times 10^{-3} \text{ mol} = 1.5 \times 10^{-3} \times 311 \text{ g}$$

$$= 0.4665 \text{ g}$$

$$\text{So, total mass of the solution} = (1000 + 0.4665) \text{ g}$$

$$= 1000.4665 \text{ g}$$

So the solution containing 0.4665 g of nalorphene weighs 1000.4665 g.

Conclusion

So, the mass of solution containing 1.5 mg of nalorphene is:

$$\frac{1000.4665 \times 1.5 \times 10^{-3}}{0.4665} \text{ g}$$

$$= 3.22 \text{ g}$$

Hence, the mass of aqueous solution required is 3.22 g. //

Discrepancy

Note: This answer differs slightly from the one given in the NCERT textbook.

Q

30

Amount of benzoic acid for a molar solution in methanol

((

Calculate the amount of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) required for preparing 250 mL of 0.15 M solution in methanol.

Given

A 0.15 M solution of benzoic acid in methanol means 1000 mL of solution contains 0.15 mol of benzoic acid.

Substitute

So, 250 mL of solution contains = $\frac{0.15 \times 250}{1000}$ mol of benzoic acid = 0.0375 mol of benzoic acid

Key formula

Molar mass of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) = $7 \times 12 + 6 \times 1 + 2 \times 16 = 122 \text{ g mol}^{-1}$



Hence, required benzoic acid = $0.0375 \text{ mol} \times 122 \text{ g mol}^{-1} = 4.575 \text{ g}$ //

Q

31

Trend in freezing point depression across acetic / trichloroacetic / trifluoroacetic acid



The depression in freezing point of water observed for the same amount of acetic acid, trichloroacetic acid and trifluoroacetic acid increases in the order given above.

Explain briefly.

Conclusion

Acetic acid

Trichloroacetic acid Trifluoroacetic acid

CH_3COOH (acetic acid), CCl_3COOH (trichloroacetic acid), CF_3COOH (trifluoroacetic acid)

Among H, Cl, and F, H is the least electronegative and F is the most electronegative. So F pulls electrons towards itself more strongly than Cl and H. This means trifluoroacetic acid loses H^+ ions most easily, so it ionizes the most. The more ions a solution has, the greater the depression in freezing point. So the freezing point depression increases in this order:

Acetic acid < trichloroacetic acid < trifluoroacetic acid

Q

32

Freezing
point
depression
from a weak
acid with
dissociation



Calculate the depression in the freezing point of water when 10 g of $\text{CH}_3\text{CH}_2\text{CHClCOOH}$ is added to 250 g of water. $K_a = 1.4 \times 10^{-3}$, $K_f = 1.86 \text{ K kg mol}^{-1}$.

Given

Molar mass of $\text{CH}_3\text{CH}_2\text{CHClCOOH} = 15+14+13+35.5+12+16+16+1$

$$= 122.5 \text{ g mol}^{-1}$$

$\therefore$ No. of moles present in 10 g of

$$\text{CH}_3\text{CH}_2\text{CHClCOOH} = \frac{10 \text{ g}}{122.5 \text{ g mol}^{-1}}$$

$$= 0.0816 \text{ mol}$$

It is given that 10 g of is added to 250 g of water.

$\text{CH}_3\text{CH}_2\text{CHClCOOH}$

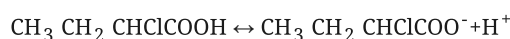
$$\therefore \text{Molality of the solution, } = \frac{0.0816}{250} \times 1000$$

$$= 0.3264 \text{ mol kg}^{-1}$$

Substitute

Let α be the degree of dissociation of $\text{CH}_3\text{CH}_2\text{CHClCOOH}$.

$\text{CH}_3\text{CH}_2\text{CHClCOOH}$ undergoes dissociation according to the following equation:



Initial conc.	C mol L ⁻¹	0	0
At equilibrium	C(1- α)	C α	C α

Key formula



$$\begin{aligned} \therefore K_a &= \frac{C \alpha \cdot C \alpha}{C(1-\alpha)} \\ &= \frac{C \alpha^2}{1-\alpha} \end{aligned}$$

Substitute

Since α is very small with respect to 1, $1-\alpha \approx 1$

Now,

$$K_a = \frac{C \alpha^2}{1}$$

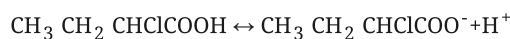
$$\Rightarrow K_a = C \alpha^2$$

$$\Rightarrow \alpha = \sqrt{\frac{K_a}{C}}$$

$$= \sqrt{\frac{1.4 \times 10^{-3}}{0.3264}} \quad (\because K_a = 1.4 \times 10^{-3})$$

$$= 0.0655$$

Again,



Initial moles	1	0	0
At equilibrium	1- α	α	α

Total moles of equilibrium = $1-\alpha+\alpha+\alpha$

$$= 1+\alpha$$

$$\begin{aligned}\therefore i &= \frac{1+\alpha}{1} \\ &= 1+\alpha \\ &= 1+0.0655 \\ &= 1.0655\end{aligned}$$

Conclusion

Hence, the depression in the freezing point of water is given as:

$$\begin{aligned}\Delta T_f &= i \cdot K_f m \\ &= 1.0655 \times 1.86 \text{ K kg mol}^{-1} \times 0.3264 \text{ mol kg}^{-1} \\ &= 0.65 \text{ K}\end{aligned}$$

$$\Delta T_f = 0.65 \text{ K} //$$

Q

33

Van't Hoff factor and dissociation constant of fluoroacetic acid



19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.0°C . Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.

Given

It is given that:

$$w_1 = 500 \text{ g}$$

$$w_2 = 19.5 \text{ g}$$

$$K_f = 1.86 \text{ K kg mol}^{-1}$$

$$\Delta T_f = 1 \text{ K}$$

Substitute

We know that:

$$\begin{aligned}M_2 &= \frac{K_f \times w_2 \times 1000}{\Delta T_f \times w_1} \\ &= \frac{1.86 \text{ K kg mol}^{-1} \times 19.5 \text{ g} \times 1000 \text{ g kg}^{-1}}{500 \text{ g} \times 1 \text{ K}} \\ &= 72.54 \text{ g mol}^{-1}\end{aligned}$$

Therefore, observed molar mass of CH_2FCOOH , $(M_2)_{\text{obs}} = 72.54 \text{ g mol}^{-1}$

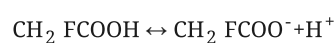
The calculated molar mass of CH_2FCOOH

$$\begin{aligned}(M_2)_{\text{cal}} &= 14+19+12+16+16+1 \\ &= 78 \text{ g mol}^{-1}\end{aligned}$$

Therefore, van't Hoff factor, is:

$$\begin{aligned}&= \frac{78 \text{ g mol}^{-1}}{72.54 \text{ g mol}^{-1}} \\ &= 1.0753\end{aligned}$$

Let α be the degree of dissociation of CH_2FCOOH



Initial conc.

C mol L⁻¹

0

0

At equilibrium	$C(1-\alpha)$	$C\alpha$	$C\alpha$	Total = $C(1+\alpha)$
----------------	---------------	-----------	-----------	-----------------------

$$\therefore i = \frac{C(1+\alpha)}{C}$$

$$\Rightarrow i = 1+\alpha$$

$$\Rightarrow \alpha = i-1$$

$$= 1.0753-1$$

$$= 0.0753$$

Key formula



Now, the value of K_a is given as:

$$K_a = \frac{[\text{CH}_2\text{FCOO}^-][\text{H}^+]}{[\text{CH}_2\text{FCOOH}]}$$

$$= \frac{C\alpha \cdot C\alpha}{C(1-\alpha)}$$

$$= \frac{C\alpha^2}{1-\alpha}$$

Substitute

Taking the volume of the solution as 500 mL, we have the concentration:

$$C = \frac{\frac{19.5}{78}}{500} \times 1000 \text{ M}$$

$$= 0.5 \text{ M}$$

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

Therefore,

$$= \frac{0.5 \times (0.0753)^2}{1-0.0753}$$

$$= \frac{0.5 \times 0.00567}{0.9247}$$

$$= 0.00307 \text{ (approximately)}$$

$$= 3.07 \times 10^{-3}$$

Van't Hoff factor $i = 1.0753$; dissociation constant $K_a = 3.07 \times 10^{-3}$.



Q

34

Vapour pressure of water lowered by glucose

(())

Vapour pressure of water at 293 K is 17.535 mm Hg . Calculate the vapour pressure of water at 293 K when 25 g of glucose is dissolved in 450 g of water.

Given



Vapour pressure of water, $p_1^0 = 17.535 \text{ mm of Hg}$

Mass of glucose, $w_2 = 25 \text{ g}$

Mass of water, $w_1 = 450 \text{ g}$

We know that,

Molar mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), $M_2 = 6 \times 12 + 12 \times 1 + 6 \times 16 = 180 \text{ g mol}^{-1}$

Molar mass of water, $M_1 = 18 \text{ g mol}^{-1}$

$$\text{Then, number of moles of glucose, } n_2 = \frac{25}{180 \text{ g mol}^{-1}}$$

$$= 0.139 \text{ mol}$$

And, number of moles of water,

$$n_1 = \frac{450 \text{ g}}{18 \text{ g mol}^{-1}}$$

$$= 25 \text{ mol}$$

Substitute

We know that,

$$\frac{p_1^0 - p_1}{p_1^0} = \frac{n_1}{n_2 + n_1}$$

$$\Rightarrow \frac{17.535 - p_1}{17.535} = \frac{0.139}{0.139 + 25}$$

$$\Rightarrow 17.535 - p_1 = \frac{0.139 \times 17.535}{25.139}$$

$$\Rightarrow 17.535 - p_1 = 0.097$$

$$\Rightarrow p_1 = 17.44 \text{ mm of Hg}$$

Hence, the vapour pressure of water is 17.44 mm of Hg. ///

Q

35

Solubility of methane in benzene from Henry's law constant



Henry's law constant for the molality of methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg.

Given

Here, $p = 760$

mm Hg $k_H = 4.27 \times 10^5$

mm Hg

Substitute

According to

Henry's law,

$$p = k_H X$$

$$\Rightarrow x = \frac{p}{k_H}$$

$$= \frac{760 \text{ mm Hg}}{4.27 \times 10^5 \text{ mm Hg}}$$

$$= 177.99 \times 10^{-5}$$

$$= 178 \times 10^{-5} \text{ (approximately)}$$

Hence, the mole fraction of methane in benzene is 178×10^{-5} . ///

Q**36**

Vapour pressures of pure liquids A and B from a mixture



100 g of liquid A (molar mass 140 g mol^{-1}) was dissolved in 1000 g of liquid B (molar mass 180 g mol^{-1}). The vapour pressure of pure liquid B was found to be 500 torr. Calculate the vapour pressure of pure liquid A and its vapour pressure in the solution if the total vapour pressure of the solution is 475 Torr.

Given

$$\text{Number of moles of liquid A, } n_A = \frac{100}{140} \text{ mol} = 0.714 \text{ mol}$$

$$\text{Number of moles of liquid B, } n_B = \frac{1000}{180} \text{ mol} = 5.556 \text{ mol}$$

$$\text{Then, mole fraction of A, } x_A = \frac{n_A}{n_A + n_B}$$

$$= \frac{0.714}{0.714 + 5.556}$$

$$= 0.114$$

$$\text{And, mole fraction of B, } x_B = 1 - 0.114$$

$$= 0.886$$

$$\text{Vapour pressure of pure liquid B, } p_B^0 = 500 \text{ torr}$$

Substitute

Therefore, vapour pressure of liquid B in the solution,

$$p_B = p_B^0 x_B$$

$$= 500 \times 0.886$$

$$= 443 \text{ torr}$$

$$\text{Total vapour pressure of the solution, } p_{\text{total}} = 475 \text{ torr}$$

$\therefore$ Vapour pressure of liquid A in the solution,

$$p_A = p_{\text{total}} - p_B = 475 - 443$$

$$= 32 \text{ torr}$$

Conclusion

Now,

$$p_A = p_A^0 x_A$$

$$\Rightarrow p_A^0 = \frac{p_A}{x_A}$$

$$= \frac{32}{0.114}$$

$$= 280.7 \text{ torr}$$

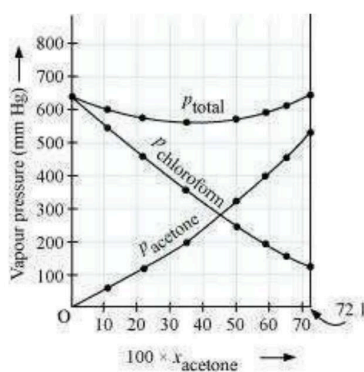
Hence, the vapour pressure of pure liquid A is 280.7 torr. Vapour pressure of liquid A in the solution, $p_A = 32 \text{ torr}$.

//

Q

37

Plotting
vapour
pressure vs
composition
for acetone-
chloroform



Plot of p_{total} , $p_{\text{chloroform}}$ and p_{acetone} as a function of x_{acetone} , referenced in the answer as "the graph".

Vapour pressures of pure acetone and chloroform at 328 K are 741.8 mm Hg and 632.8 mm Hg respectively. Assuming that they form ideal solution over the entire range of composition, plot p_{total} , $p_{\text{chloroform}}$, and p_{acetone} as a function of x_{acetone} . The experimental data observed for different compositions of mixture is:

$100 \times x_{\text{acetone}}$	0	11.8	23.4	36.0	50.8	58.2	64.5	72.1
p_{acetone} / mm Hg	0	54.9	110.1	202.4	322.7	405.9	454.1	521.1
$p_{\text{chloroform}}$ / mm Hg	632.8	548.1	469.4	359.7	257.7	193.6	161.2	120.7

Plot this data also on the same graph paper.

Indicate whether it has positive deviation or negative deviation from the ideal solution.

Conclusion

From the question, we have the following data

It can be observed from the graph that the plot for the p_{total} of the solution curves downwards. Therefore, the solution shows negative deviation from the ideal behaviour.

///

$100 \times x_{\text{acetone}}$	0	11.8	23.4	36.0	50.8	58.2	64.5	72.1
Pacetone / mm Hg	0	54.9	110.1	202.4	322.7	405.9	454.1	521.1
Pchloroform / mm Hg	632.8	548.1	469.4	359.7	257.7	193.6	161.2	120.7
p_{total} (mm Hg)	632.8	603.0	579.5	562.1	580.4	599.5	615.3	641.8

Q

38

Mole fraction
of benzene in
vapour phase
(benzene-
toluene)

(())

Benzene and toluene form ideal solution over the entire range of composition. The vapour pressure of pure benzene and toluene at 300 K are 50.71 mm Hg and 32.06 mm Hg respectively. Calculate the mole fraction of benzene in vapour phase if 80 g of benzene is mixed with 100 g of toluene.

Given

Molar mass of benzene (C_6H_6) = $6 \times 12 + 6 \times 1$
= 78 g mol^{-1}

Molar mass of toluene ($\text{C}_6\text{H}_5\text{CH}_3$) = $7 \times 12 + 8 \times 1$
= 92 g mol^{-1}

Now, no. of moles present in 80 g of benzene

And, no. of moles present in 100 g of toluene

$$= \frac{80}{78} \text{ mol} = 1.026 \text{ mol}$$

$$= \frac{100}{92} \text{ mol} = 1.087 \text{ mol}$$

∴ Mole fraction of benzene, X_b

And, mole fraction of toluene,

$$= \frac{1.026}{1.026+1.087} = 0.486$$

$$x_t = 1-0.486 = 0.514$$

It is given that vapour pressure of pure benzene,

$$p_h^0 = 50.71 \text{ mm Hg}$$

And, vapour pressure of pure toluene,

$$p_t^0 = 32.06 \text{ mm Hg}$$

Substitute

Therefore, partial vapour pressure of benzene,

$$= 0.486 \times 50.71$$

$$= 24.645 \text{ mm Hg}$$

$$p_b = x_b \times p_b^0$$

And, partial vapour pressure of toluene, $p_t = x_t \times p_t^0$

$$= 0.514 \times 32.06$$

$$= 16.479 \text{ mm Hg}$$

Key formula



$$\frac{p_b}{p_b + p_t}$$

Substitute

$$= \frac{24.645}{24.645+16.479}$$

$$= \frac{24.645}{41.124}$$

Conclusion

$$= 0.599$$

$$= 0.6$$

Mole fraction of benzene in vapour phase ≈ 0.6 .

///

Q

39

Composition
of dissolved
O₂ and N₂ in
water from
air



The air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298 K if the Henry's law constants for oxygen and nitrogen at 298 K are 3.30×10^7 mm and 6.51×10^7 mm respectively, calculate the composition of these gases in water.

Given

Percentage of oxygen (O₂) in air = 20 %

Percentage of nitrogen (N₂) in air = 79 %

Water is in equilibrium with air at a total pressure of 10 atm. That is, (10 × 760) mm Hg = 7600 mm Hg

Therefore,

$$\text{Partial pressure of oxygen, } p_{\text{O}_2} = \frac{20}{100} \times 7600 \text{ mm Hg}$$

$$= 1520 \text{ mm Hg}$$

$$\text{Partial pressure of nitrogen, } p_{\text{N}_2} = \frac{79}{100} \times 7600 \text{ mmHg}$$

$$= 6004 \text{ mmHg}$$

Key formula

Now, according to Henry's law:

$$p = K_H \cdot x$$



Substitute

For oxygen:

$$p_{\text{O}_2} = K_H \cdot x_{\text{O}_2}$$

$$\Rightarrow x_{\text{O}_2} = \frac{p_{\text{O}_2}}{K_H}$$

$$= \frac{1520 \text{ mm Hg}}{3.30 \times 10^7 \text{ mm Hg}} \quad (\text{Given } K_H = 3.30 \times 10^7 \text{ mm Hg})$$

$$= 4.61 \times 10^{-5}$$

For nitrogen:

$$p_{\text{N}_2} = K_H \cdot x_{\text{N}_2}$$

$$\Rightarrow x_{\text{N}_2} = \frac{p_{\text{N}_2}}{K_H}$$

$$= \frac{6004 \text{ mm Hg}}{6.51 \times 10^7 \text{ mm Hg}}$$

$$= 9.22 \times 10^{-5}$$

Hence, the mole fractions of oxygen and nitrogen in water are 4.61×10^{-5} and 9.22×10^{-5} respectively.

//

Q

40

Amount of
CaCl₂ for a
given
osmotic
pressure



Determine the amount of CaCl₂ (i = 2.47) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27° C.

Given

We know that,

$$\pi = i \frac{n}{V} R T$$

$$\Rightarrow \pi = i \frac{w}{M V} R T$$

$$\Rightarrow w = \frac{\pi M V}{i R T}$$

$$\pi = 0.75 \text{ atm}$$

$$V = 2.5 \text{ L}$$

$$i = 2.47$$

$$T = (27+273) \text{ K} = 300 \text{ K}$$

Here,

$$R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

$$M = 1 \times 40 + 2 \times 35.5$$

$$= 111 \text{ g mol}^{-1}$$

Substitute

$$\begin{aligned} \text{Therefore, } w &= \frac{0.75 \times 111 \times 2.5}{2.47 \times 0.0821 \times 300} \\ &= 3.42 \text{ g} \end{aligned}$$

Hence, the required amount of CaCl₂ is 3.42 g.

///

Q

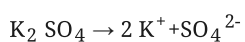
41

Osmotic
pressure of a
K₂SO₄
solution



Determine the osmotic pressure of a solution prepared by dissolving 25 mg of K₂SO₄ in 2 litre of water at 25° C, assuming that it is completely dissociated.

Given

When K₂SO₄ is dissolved in water, K⁺ and SO₄²⁻ ions are produced.

Total number of ions produced = 3

$$\therefore i = 3 \text{ Given, } w = 25 \text{ mg} = 0.025$$

g

$$V = 2 \text{ L}$$

$$T = 25^\circ \text{ C} = (25+273) \text{ K} = 298 \text{ K Also,}$$

we know that:

$$R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

$$M = (2 \times 39) + (1 \times 32) + (4 \times 16) = 174 \text{ g mol}^{-1}$$

Substitute

Applying the following relation,

$$\pi = i \frac{n}{V} R T$$

///

$$= i \frac{w}{M} \frac{1}{v} R T$$

$$\pi = 5.27 \times 10^{-3} \text{ atm}$$

$$= 3 \times \frac{0.025}{174} \times \frac{1}{2} \times 0.0821 \times 298$$